

MOSFET Characterization Suite

TSMC 180nm BSIM3 | LTspice 26 | May 2026

1. Purpose

Every analog design decision, transistor sizing, bias current, gain, bandwidth, depends on knowing how the transistor behaves in the target process. This project directly sweeps and measures NMOS and PMOS transistors in the TSMC 180nm BSIM3 model. The output is a personal reference that replaces guesswork with reasoning in every future project.

2. Why This Project Exists

Textbook Level-1 SPICE models are too crude for real analog design. A TSMC 180nm transistor exhibits velocity saturation, mobility degradation, subthreshold conduction, and channel-length modulation, none of which Level-1 captures. Without characterizing the real model first, a designer cannot:

- Predict g_m at a given bias current without trial-and-error simulation*
- Choose between low-power (high g_m/I_d) and high-speed (low g_m/I_d) operating points*
- Estimate output resistance r_o or intrinsic gain $A_v = g_m \times r_o$*
- Explain why PMOS needs $\sim 2\times$ the width of NMOS for matched transconductance*

3. What Was Done

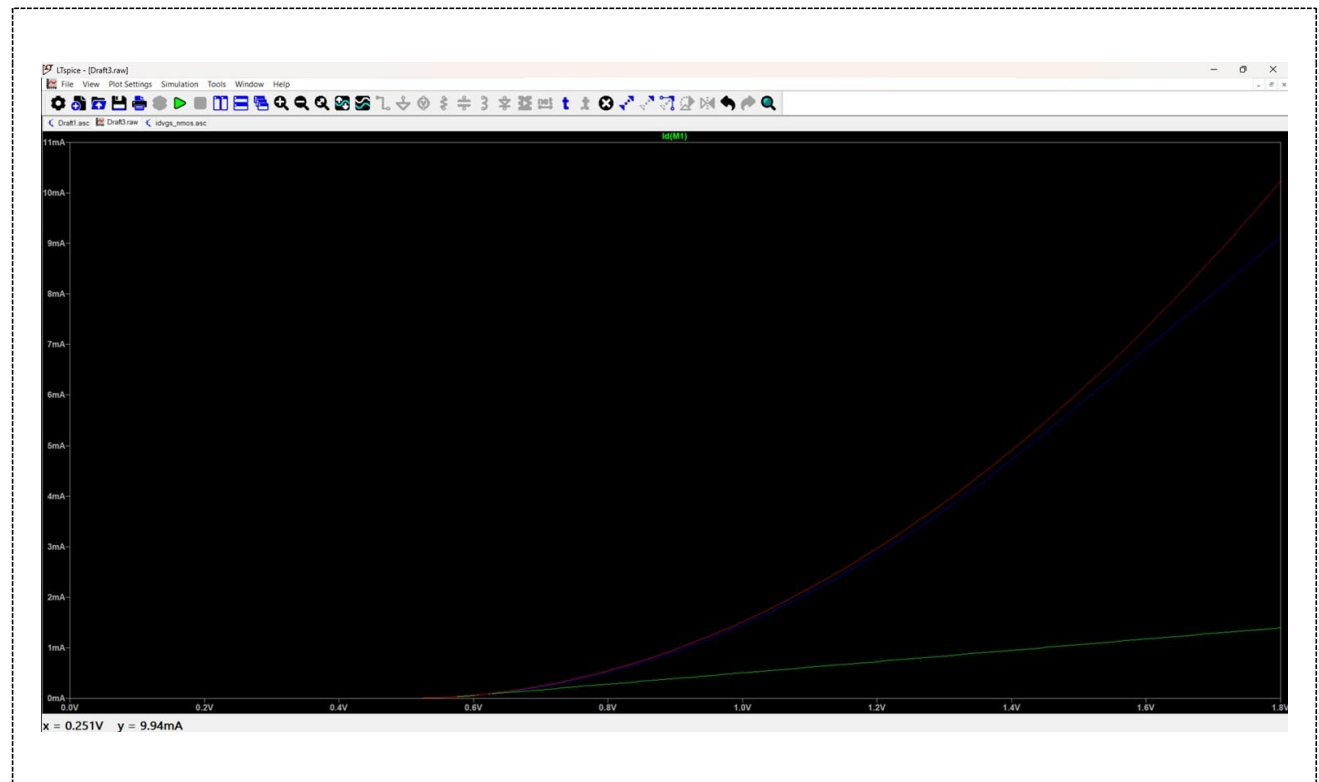
Five simulation sweeps were performed for both NMOS and PMOS. All schematics use $W = 10\text{ }\mu\text{m}$, $L = 0.18\text{ }\mu\text{m}$, $V_{DD} = 1.8\text{ V}$, TSMC 180nm BSIM3v3 model (MOSIS wafer run T14B).

Sweep	What It Shows	Key Finding
I_d vs V_{gs}	Turn-on curve, threshold voltage	$V_{th} \sim 0.35\text{ V}$ (NMOS), 0.42 V (PMOS)
I_d vs V_{ds}	Triode / saturation boundary	Knee at $V_{ds} = V_{ov} = V_{gs} - V_{th}$
g_m vs V_{gs}	Transconductance across bias range	NMOS peak 4.0 mA/V ; PMOS 2.2 mA/V
g_m/I_d vs V_{gs}	Efficiency: low-power vs high-speed	Range 27 (sub- V_{th}) to 4 V^{-1} (strong inv.)
.op extraction	Exact small-signal parameters at bias	Reference point: $V_{gs} = V_{ds} = 0.9\text{ V}$

4. Simulation Results

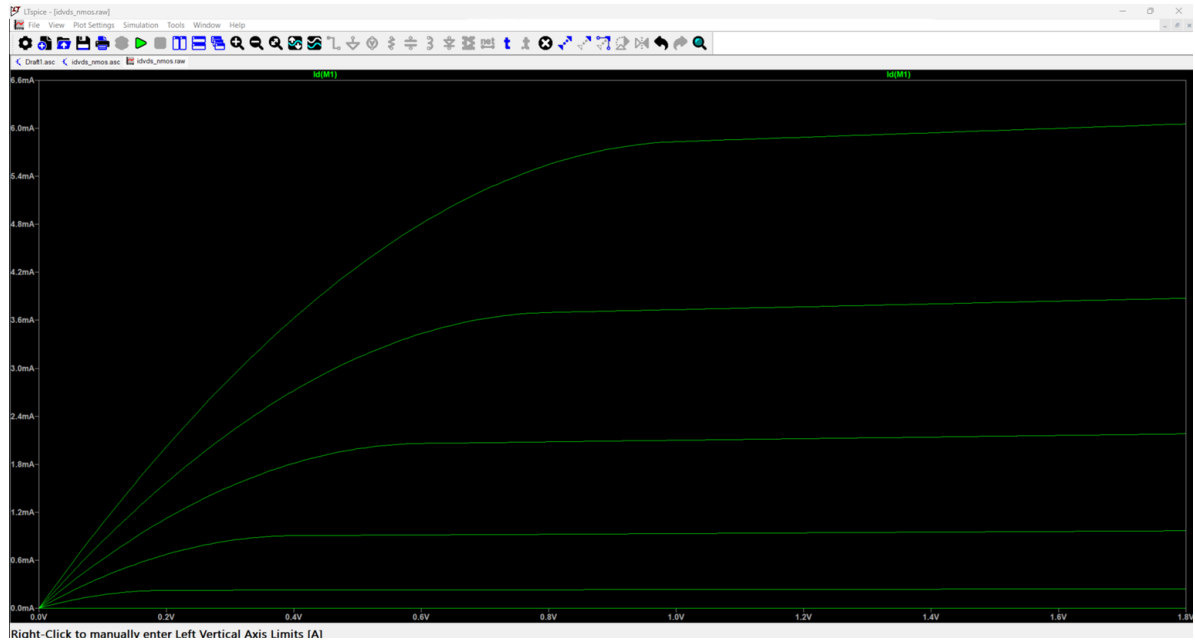
4.1 NMOS - I_d vs V_{gs}

$V_{th} \sim 0.35\text{ V}$ visible as the knee. Three curves at $V_{ds} = 0.1\text{ V}$, 0.95 V , 1.8 V .



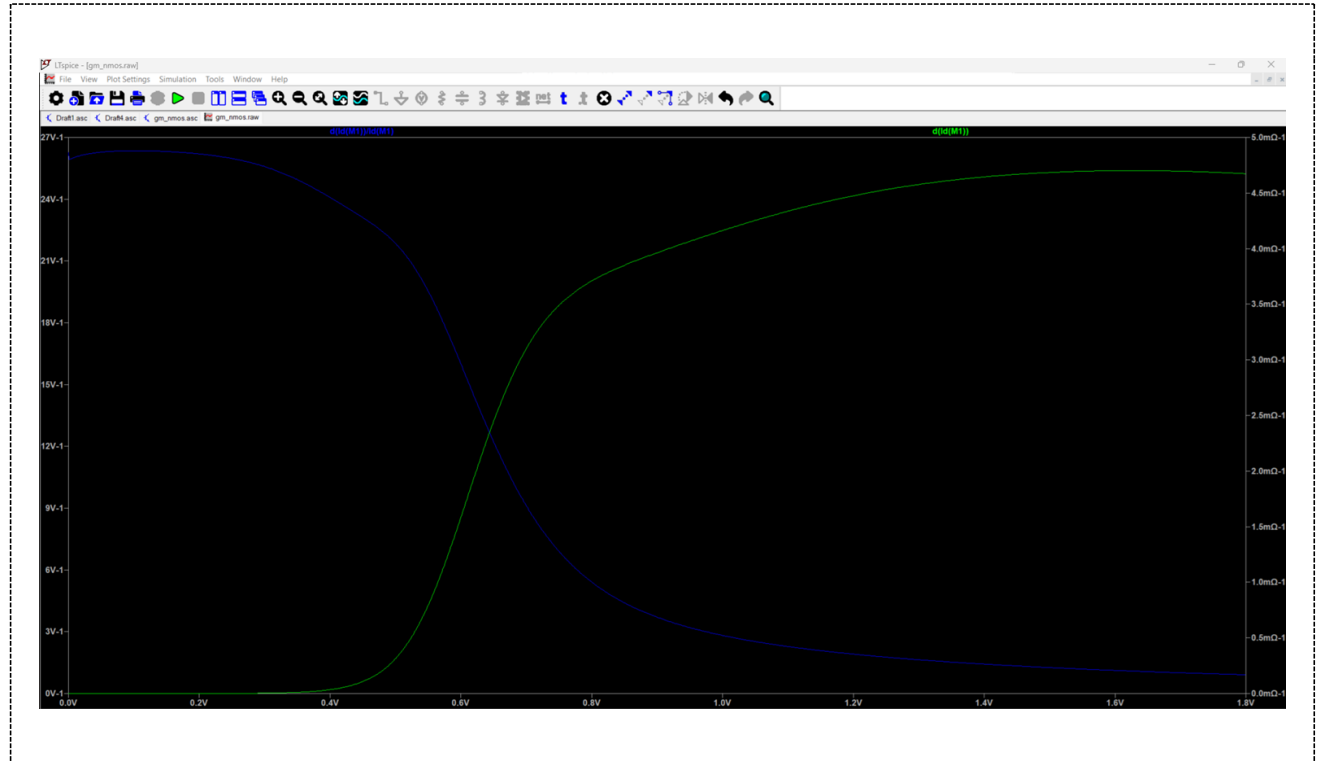
4.2 NMOS - I_d vs V_{ds}

Six curves ($V_{gs} = 0.5\text{ V}$ to 1.5 V , step 0.2 V). Triode-to-saturation knee shifts right as V_{gs} increases. Upward slope in saturation = channel-length modulation (finite r_o).



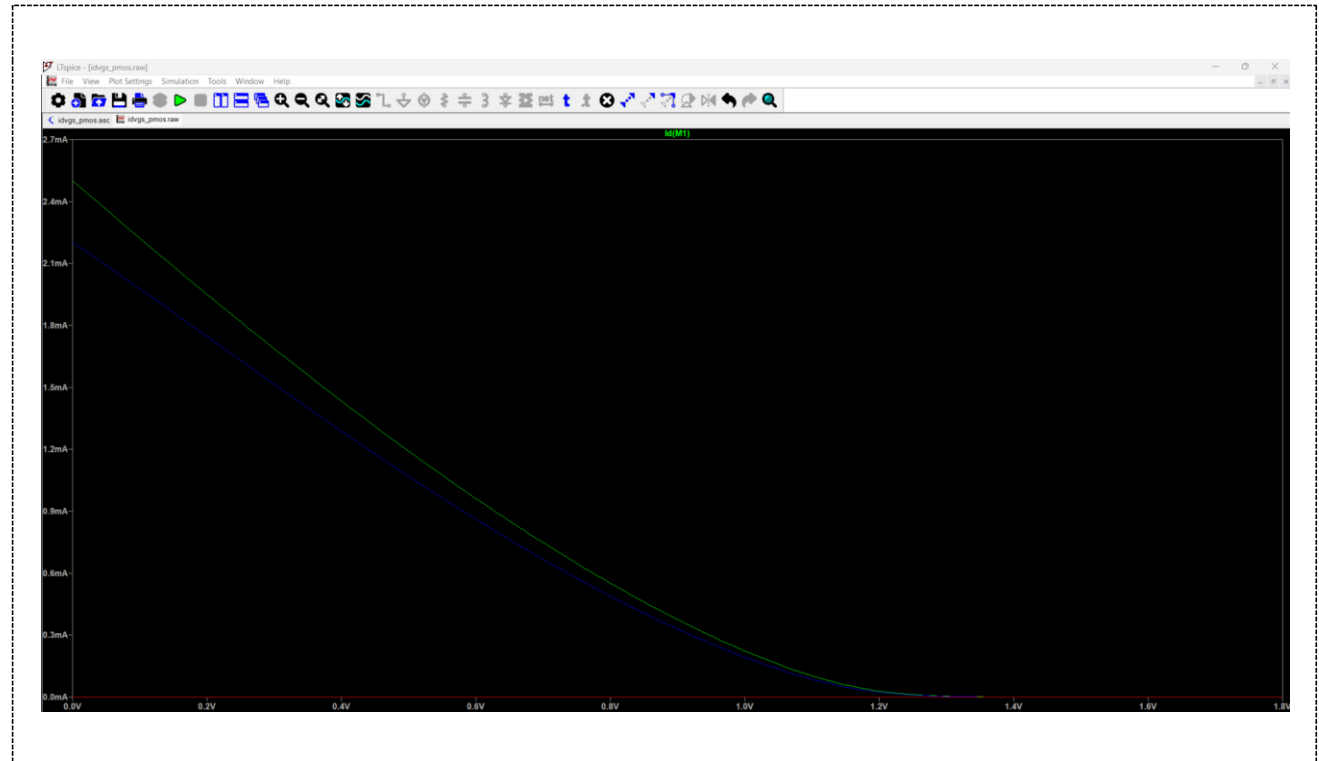
4.3 NMOS - gm and gm/Id vs Vgs

The primary design plot. gm/Id starts at $\sim 27 \text{ V}^{-1}$ in subthreshold and falls to $\sim 4 \text{ V}^{-1}$ in deep strong inversion. Used to select the bias point for every future amplifier.



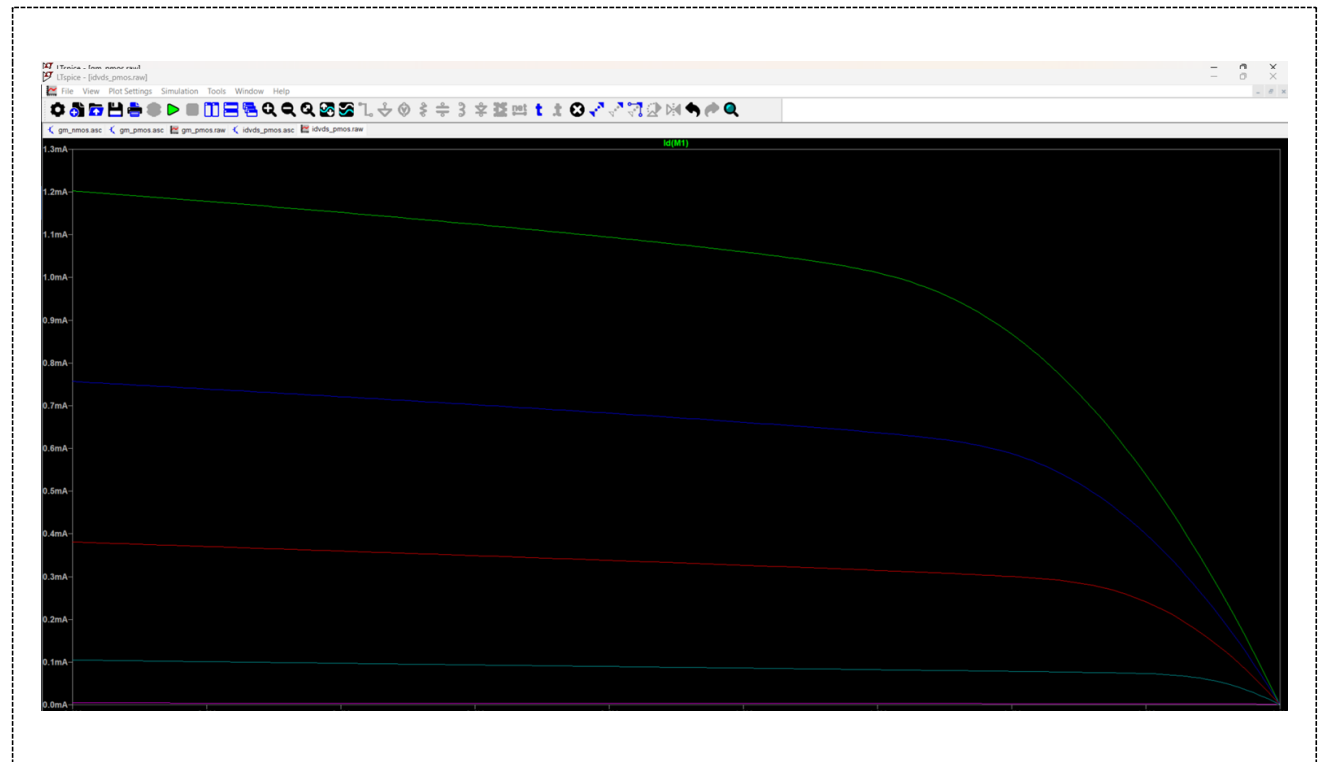
4.4 PMOS - I_d vs V_{gs}

Mirror image of NMOS: fully ON at $V_{gs} = 0$ V, turns off near $V_{gs} \sim 1.38$ V ($= V_{DD} - |V_{th}|$).



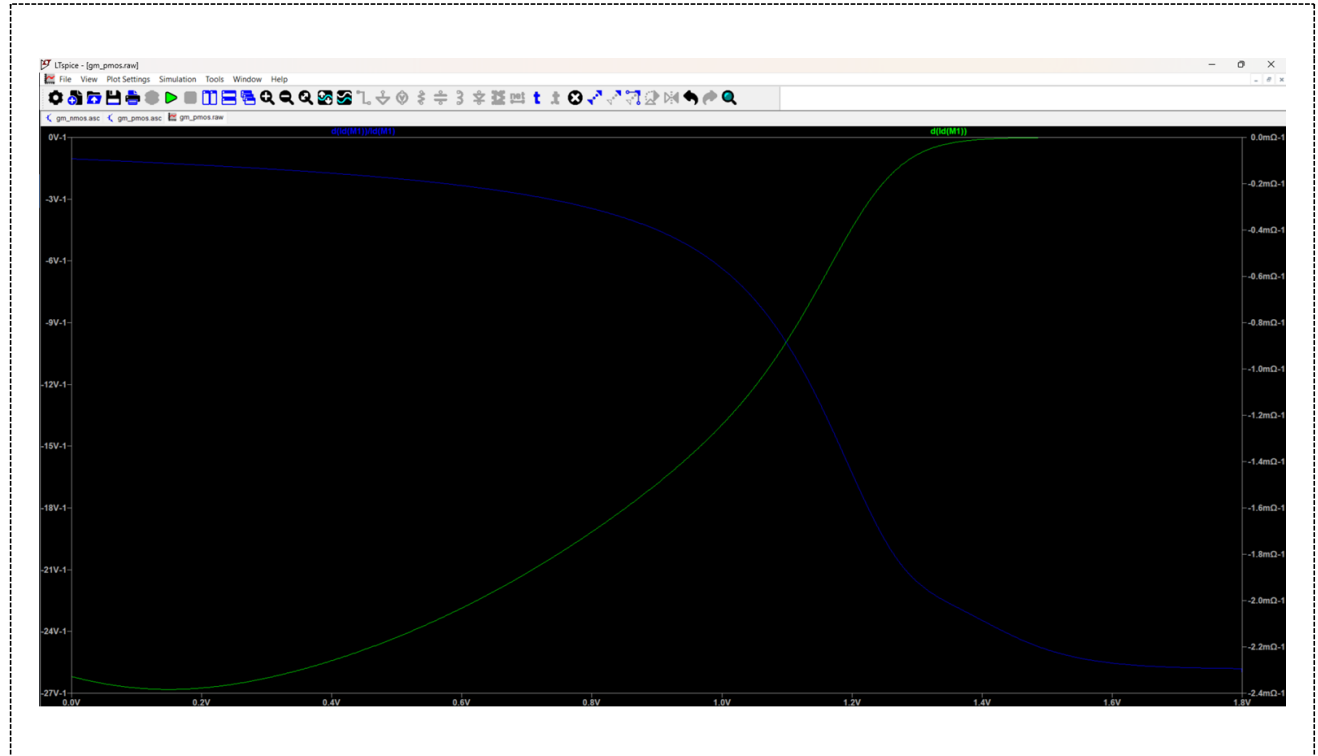
4.5 PMOS - I_d vs V_{ds}

Current is highest at low V_{ds} and reduces as V_{ds} rises toward V_{DD} .



4.6 PMOS - gm and gm/Id vs Vgs

gm is negative (current decreases with increasing Vgs). Peak magnitude ~2.2 mA/V, roughly 55% of NMOS, reflecting lower hole mobility.



5. Extracted Parameters - Reference Bias

Bias point: $V_{gs} = 0.9\text{ V}$, $V_{ds} = 0.9\text{ V}$, $W/L = 10\text{ }\mu\text{m} / 0.18\text{ }\mu\text{m}$, $V_{DD} = 1.8\text{ V}$.

Parameter	NMOS	PMOS
V_{th}	$\sim 0.35\text{ V}$	$\sim -0.42\text{ V}$
$V_{ov} = V_{gs} - V_{th}$	$\sim 0.55\text{ V}$	$\sim 0.48\text{ V}$
I_d	1.07 mA	-
g_m	4.0 mA/V	$\sim 2.2\text{ mA/V}$
g_m/I_d	3.79 V^{-1}	4.5 V^{-1}
$r_o = 1/g_{ds}$	$\sim 2.86\text{ k}\Omega$	-
<i>Intrinsic gain ($g_m \times r_o$)</i>	$\sim 11.4\text{ V/V}$	-

6. Key Design Insights

- $V_{th} \sim 0.35 \text{ V}$ (NMOS) and 0.42 V (PMOS). $V_{ds} > V_{ov}$ to confirm saturation.
- $g_m/I_d = 15\text{-}20 \text{ V}^{-1} = \text{low-power / low-noise region}$. $g_m/I_d = 5\text{-}10 \text{ V}^{-1} = \text{high-speed region}$. This is the primary bias selection knob.
- PMOS $g_m \sim 55\%$ of NMOS g_m at same current. PMOS needs $\sim 2\times$ wider W for matched g_m in a differential pair.
- Intrinsic gain at $L = 0.18 \text{ }\mu\text{m}$ is only $\sim 11 \text{ V/V}$ due to low r_o . Increasing L to $0.5\text{-}1 \text{ }\mu\text{m}$ raises gain to $60\text{-}160 \text{ V/V}$. Analog gain stages always use longer L .
- Minimum L is reserved for speed-critical paths only. Never use for gain stages.

7. Project Files

File	Description
<i>gm_pmos</i>	<i>PMOS g_m and g_m/I_d vs V_{gs}</i>
<i>idvds_pmos</i>	<i>PMOS I_d vs V_{ds} sweep</i>
<i>idvgs_pmos</i>	<i>PMOS I_d vs V_{gs} sweep</i>
<i>gm_nmos</i>	<i>NMOS g_m and g_m/I_d vs V_{gs}</i>
<i>idvds_nmos</i>	<i>NMOS I_d vs V_{ds} sweep</i>
<i>idvgs_nmos</i>	<i>NMOS I_d vs V_{gs} sweep</i>